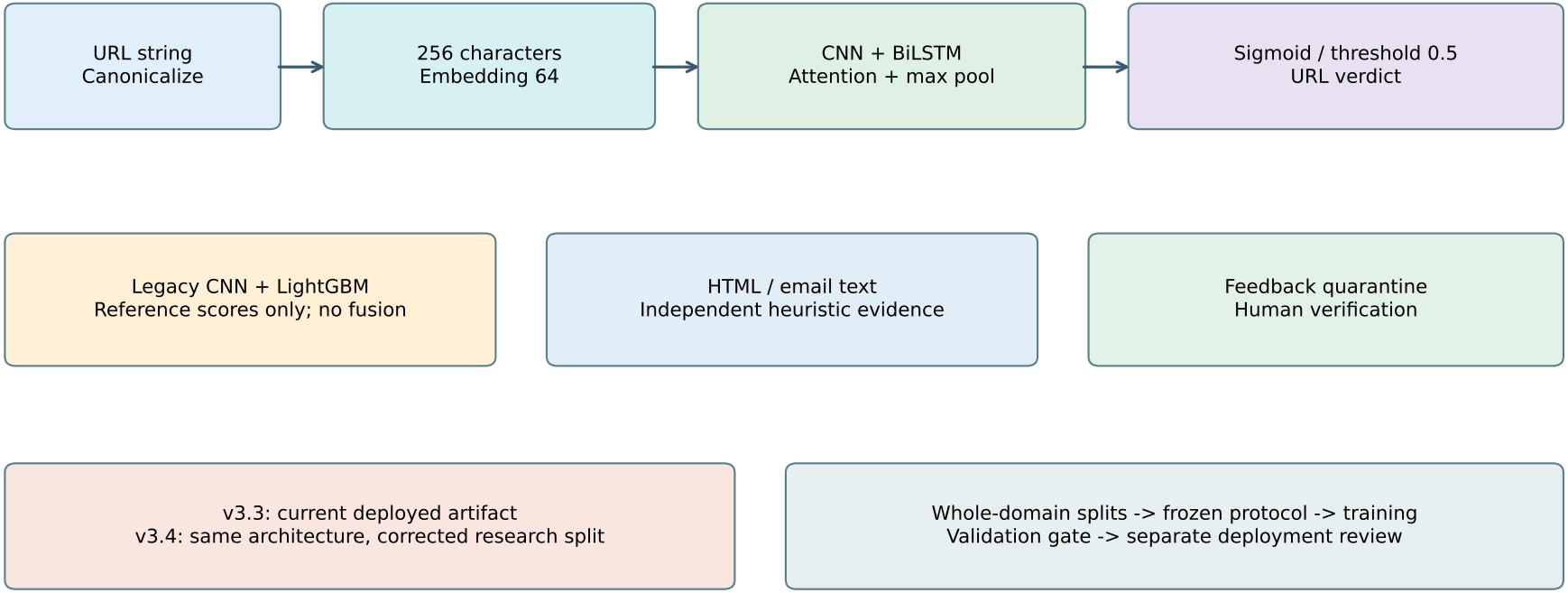
URL string
Canonicalize
256 characters
Embedding 64
CNN + BiLSTM
Attention + max pool
Sigmoid / threshold 0.5
URL verdict
Legacy CNN + LightGBM
Reference scores only; no fusion
HTML / email text
Independent heuristic evidence
Feedback quarantine
Human verification
v3.3: current deployed artifact
v3.4: same architecture, corrected research split
Whole-domain splits -> frozen protocol -> training
Validation gate -> separate deployment review